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Section A

1. (3) 3. ✓

2. (2) $\tan 60^\circ$ ✓

3. (4) 100° ✓

4. (1) 10 cm ✓

5. (4) $a = 0$ ✓

6. (4) Four decimal places. ✓

7. (3) 3 & 1 ✓

8. (2) $(0, -1)$ ✓

9. (4) No solution. ✓

10. (3) $k = \frac{9}{2}$ ✓

11. (1) $k \neq -3$ ✓

12. (1) 8 ✓

13. (1) similar. ✓

14. (1) $x^3 y^4 z^4$ ✓

15. (3) $\frac{x^2}{2} - \frac{1}{2} - 6$ ✓

16. (3) intersecting or coincident. ✓

17. (1) ~~Parallel~~ Parallel ✓

18. (4) 16:81. ✓

19. (3) Assertion is true but Reason (R) is false. ✓

20. (1) Both A & R are true & R is the correct explanation of A. ✓

section - B

21. $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

$\Rightarrow 720 + 5$

$\Rightarrow 725$

as the no. ends with 5 that means 5 is a factor of 725
hence it has more than 2 factors.

hence $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ is a composite no.

22. $4x + 6y = 7$ to $kx + 18y = 21$ (1)

- in eq (1)

now, for let $a_1 = 4, b_1 = 6, c_1 = 7$

In eq (2) $a_2 = k, b_2 = 18, c_2 = 21$

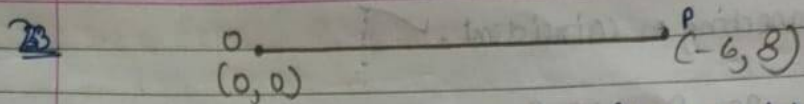
In order to obtain infinitely many solutions

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$\Rightarrow \frac{4}{k} = \frac{6}{18} = \frac{7}{21}$

taking first 2 pair

$\Rightarrow \underline{k = 12}$



using distance formula, for point (point to origin)

$$\Rightarrow \overline{OP} = \sqrt{x^2 + y^2}$$

$$\Rightarrow \overline{OP} = \sqrt{(-6)^2 + (8)^2}$$

$$\Rightarrow \overline{OP} = \sqrt{36 + 64} = \sqrt{100}$$

$$\underline{\underline{\overline{OP} = 10 \text{ units}}}$$

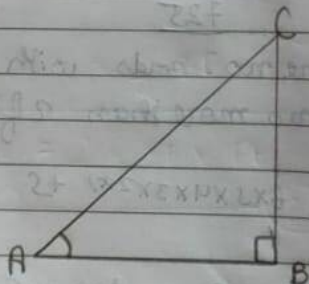
24. Given

$$\sin 3A = \cos(A - 10^\circ)$$

$$3A = \text{acute angle} = 90^\circ$$

to find: $\angle A$

now,



$$\Rightarrow \cos(A - 10^\circ)$$

$$\Rightarrow \sin 3A = \cos(A - 10^\circ)$$

$$\Rightarrow \cos(90^\circ - 3A) = \cos(A - 10^\circ) \quad [\because \sin \theta = \cos(90^\circ - \theta)]$$

$$\Rightarrow 90^\circ - 3A = A - 10^\circ$$

$$\Rightarrow 74A = 100^\circ$$

$$A = \frac{100^\circ}{4} = 25^\circ$$

hence, $\underline{\underline{\angle A = 25^\circ}}$

25. Given
 $AB \parallel CD$

to Prove: $\triangle AOB \sim \triangle DOC$

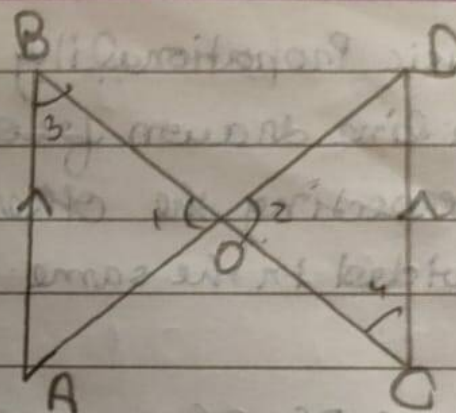
Soln In $\triangle AOB$ & $\triangle DOC$

(i) $\angle 1 = \angle 2$ (Vertically opposite)

(ii) $\angle 3 = \angle 4$ (As $AB \parallel CD$, so, Alternate interior angles)

So by AA Similarity criterion

$\triangle AOB \sim \triangle DOC$



26. to Prove

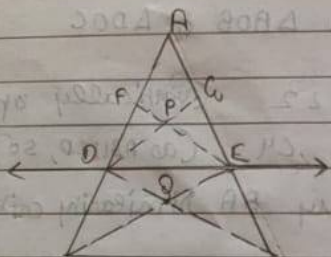
$$\frac{\cos(90^\circ - \theta)}{1 + \sin(90^\circ - \theta)} + \frac{1 + \sin(90^\circ - \theta)}{\cos(90^\circ - \theta)} = 2 \operatorname{cosec} \theta$$

32. Basic Proportionality Theorem.

If a line drawn from parallel to one side of Δ intersecting the other 2 sides, then the other 2 sides are divided in the same ratio.

Proof: $\Rightarrow$ Given: $DE \parallel BC$

To Prove: $\frac{AD}{DB} = \frac{AE}{EC}$



Proof:

Construction: let us join DC &

Point E & B . let us draw $EF \perp AD$, $EG \perp AE$.

Proof: now $\text{Ar. } (\triangle ADE) = \frac{1}{2} \times \text{base} \times \text{h} = \frac{1}{2} \times AD \times EF$

$\Rightarrow$ now $\text{Ar. } (\triangle BDE) = \frac{1}{2} \times \text{base} \times \text{h} = \frac{1}{2} \times BD \times EF$

$\Rightarrow$ Now we divide eq (1) by eq (2)

$\Rightarrow \frac{\text{Ar. } (\triangle ADE)}{\text{Ar. } (\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EF}{\frac{1}{2} \times BD \times EF} = \frac{AD}{BD}$

Similarly, $\frac{\text{Ar. } (\triangle ADE)}{\text{Ar. } (\triangle CDE)} = \frac{\frac{1}{2} \times AE \times EG}{\frac{1}{2} \times EC \times EG} = \frac{AE}{EC}$

Now ΔBDE & ΔCDE are on same base & between a same line DE & BC .

So, $\text{Ar. } (\triangle BDE) = \text{Ar. } (\triangle CDE)$

from eq (3) & (5)

$$\frac{A. (\triangle ADE)}{B. (\triangle BDE)} = \frac{AE}{EC} \quad (6)$$

now from eq (6) & (3)

$$\Rightarrow \frac{AE}{EC} = \frac{AD}{BD}$$

$$\Rightarrow \frac{AD}{DB} = \frac{AE}{EC} \text{ hence proved.}$$

26 to Prove $\frac{\cos(90^\circ - \theta)}{1 + \sin(90^\circ - \theta)} + \frac{1 + \sin(90^\circ - \theta)}{\cos(90^\circ - \theta)} = 2 \operatorname{cosec} \theta$

$\Rightarrow$ LHS side

$$\Rightarrow \frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} \quad \left[\begin{array}{l} \cos(90^\circ - \theta) = \sin \theta \\ \sin(90^\circ - \theta) = \cos \theta \end{array} \right]$$

$$\Rightarrow \frac{\sin^2 \theta + (1 + \cos \theta)^2}{(1 + \cos \theta)(\sin \theta)}$$

$$\Rightarrow \frac{\sin^2 \theta + 1 + \cos^2 \theta + 2 \cos \theta}{\sin \theta (1 + \cos \theta)}$$

$$\Rightarrow \frac{1 + 1 + 2 \cos \theta}{\sin \theta (1 + \cos \theta)} \quad [\sin^2 \theta + \cos^2 \theta = 1]$$

$$\Rightarrow \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)}$$

$$\Rightarrow \frac{2}{\sin \theta} = 2 \operatorname{cosec} \theta$$

LHS = RHS
 hence proved

22 Let the Present age of Sagar be = x years
Let the "age" time be = y years

as per Given condition

$$(1) \quad (x-5) = 2(y-5)$$

$$\Rightarrow x-5 = 2y-10$$

$$\Rightarrow \underline{x-2y+5=0} \quad (1)$$

as per another Given condition

$$\Rightarrow (x+10) = (y+10)+10$$

$$\Rightarrow \underline{x=y+20} \quad (2)$$

$$\Rightarrow \underline{x-y-10=0} \quad (2)$$

now subtracting eq (2) from eq (1)

$$\Rightarrow x-2y+5 - (x-y-10) = 0$$

$$\Rightarrow 5-y = -10$$

$$\Rightarrow \underline{y=15} \quad (3)$$

now substituting value of y from eq (3) in eq (2)

$$\Rightarrow x-15-10=0$$

$$\Rightarrow \underline{x=25}$$

∴, Present age of Sagar = $x = 25$ years
Present age of Lian = $y = 15$ years.

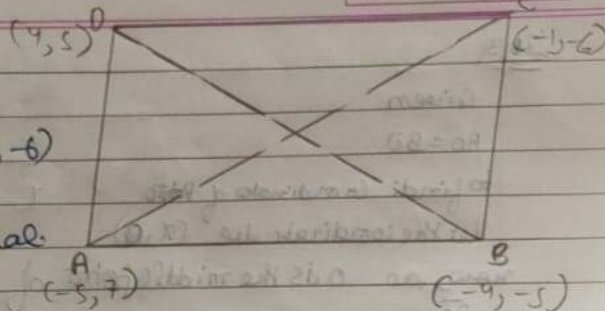
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Given

Quadrilateral ABCD

A(-5, 7), B(-4, -5), C(-1, -6)
 D(4, 5)

to find: Area of Quadrilateral ABCD



Construction: let us draw diagonals.

Proof: now, Area of $\triangle ABC = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$

here, $x_1 = -5, y_1 = 7, x_2 = -4, y_2 = -5, x_3 = -1, y_3 = -6$.

$$\therefore \text{Ar.}(\triangle ABC) = \frac{1}{2} [5(-5+6) + (-4)(-6-7) + (-1)(7+5)]$$

$$\Rightarrow \frac{1}{2} [5 + -4 \times (-13) - 12] = \frac{1}{2} [5 + 52 - 12]$$

$$\Rightarrow \frac{1}{2} [35] \text{ unit}^2 \text{ or } 17.5 \text{ unit}^2 \quad (1)$$

now similar in $\text{Ar.}(\triangle ADC) = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$

$\Rightarrow$ here $x_1 = -5, y_1 = 7, x_2 = 4, y_2 = 5, x_3 = -1, y_3 = -6$

$$\therefore \text{Ar.}(\triangle ADC) = \frac{1}{2} [-5(5+6) + 4(-6-7) - 1(7-5)]$$

$$\Rightarrow \frac{1}{2} [-55 + (-52) - 2] = \frac{1}{2} (-109) = -54.5 \text{ unit}^2 \text{ or } 54.5 \text{ unit}^2 \quad (2)$$

$$\text{now Ar. (Quadrilateral ABCD)} = \text{Ar.}(\triangle ABC) + \text{Ar.}(\triangle ADC)$$

$$= 17.5 + 54.5$$

$$\Rightarrow \text{Ar. (Quadrilateral ABCD)} = 72 \text{ unit}^2$$

30.31

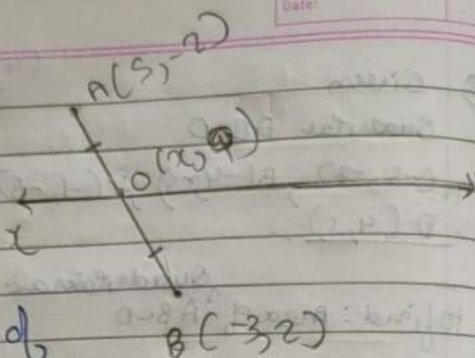
Given

$$AO = BO$$

to find: Coordinates of Point O

Let the coordinate be (x, y)

now, as O is the middle point of AB



$$\text{now, } x = \frac{(x_2 + x_1)}{2} = \frac{-3 + 5}{2} = 1$$

so the coordinates of a point on x axis which is equidistant from points A & B is $O(1, 0)$

33 Graphically

$$x - y + 1 = 0 \quad \text{--- (1)} \quad \& \quad 3x + 2y - 12 = 0 \quad \text{--- (2)}$$

in 1st eqn

$$\Rightarrow x - y = -1$$

$$x = y - 1$$

	B	A	C
x	-1	-2	1
y	0	-1	2
	(-1, 0)	(-2, -1)	(1, 2)

for 2nd eqn

$$\Rightarrow 3x + 2y = 12$$

$$3x = 12 - 2y$$

$$x = \frac{12 - 2y}{3}$$

	B	A	C
x	6	4	2
y	-3	0	3
	(6, -3)	(4, 0)	(2, 3)

now

now, let's find the area of $\triangle BAC$

$$\Rightarrow \frac{1}{2} \times b \times h = \frac{1}{2} \times 5 \times 3 = \frac{15}{2}$$

$$\text{so ar. } (\triangle BAC) = \frac{15}{2} \text{ unit}^2 \text{ or } 7.5 \text{ unit}^2$$

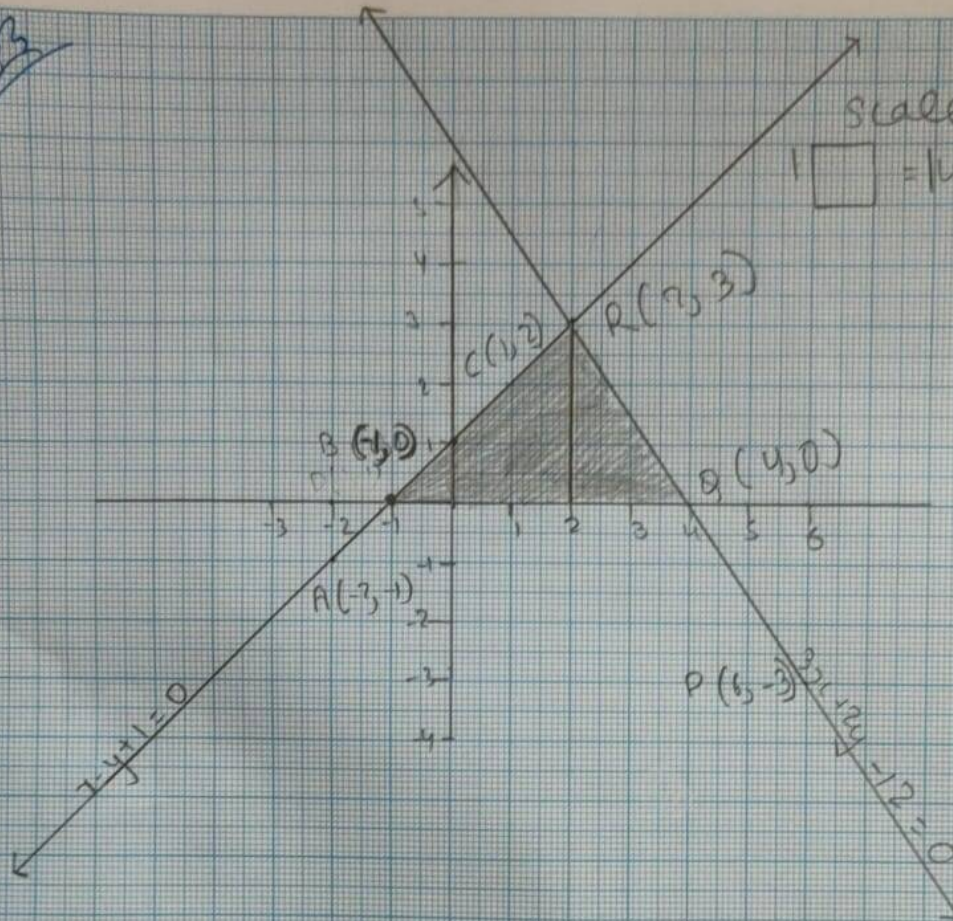
$$x^2 = \sqrt{2}x$$

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Scale:
1 $\square$ = 1 unit



34. $p(x) = 2x^3 - 4x - x^2 + 2$
 2 zeros = $\sqrt{2}, -\sqrt{2}$

$\Rightarrow \therefore x = \sqrt{2} \text{ \& } x = -\sqrt{2}$

$\Rightarrow (x + \sqrt{2})(x - \sqrt{2}) = x^2 - 2$

hence, $(x^2 - 2)$ is a factor of $p(x)$

now by Applying the division algorithm to the polynomials

$p(x) = 2x^3 - 4x - x^2 + 2 \text{ \& } x^2 - 2:$

$\Rightarrow \text{Remainder} = r = 0$

By division algorithm

$p(x) = q(x) \cdot g(x)$

$\Rightarrow \therefore 2x^3 - 4x - x^2 + 2 = (x^2 - 2)(2x - 1)$

RHS side

$= (x - \sqrt{2})(x + \sqrt{2})(2x - 1) = 0 \quad (\text{all the zeros})$

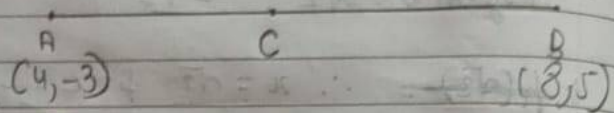
$\Rightarrow \text{if } 2x - 1 = 0$

then, $2x = 1$

$x = \frac{1}{2}$

hence, all 3 zeros of cubic poly. are $\sqrt{2}, -\sqrt{2} \text{ \& } \frac{1}{2}$.

35



Given: Point C divides AB into ratio of 3:1

$$\Rightarrow \frac{AC}{BC} = \frac{3}{1} \quad \text{--- (1)}$$

to find: Coordinates of Point C

now we using section formula let's find the coordinate of point C.

$$\Rightarrow x = \frac{3 \times 8 + 1 \times 4}{3+1} = \frac{24+4}{4} = \frac{28}{4} = 7$$

$$\underline{x = 7}$$

$$\text{now } y = \frac{3 \times 5 + 1 \times (-3)}{3+1} = \frac{15-3}{4} = \frac{12}{4} = 3$$

$$\underline{y = 3}$$

So, the coordinates of point which divides AB into the ratio of 3:1 are C(7, 3).

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36 (i) (2) (0, 12) ✓

(ii) (1) (3, 5) ✓

(iii) (3) 5 m ✓

37 (i) (3) Parabola ✓

(ii) (3) 0 ✓

(iii) (9) -3, 2 ✓

38 (i) 48 cm² ✓

(ii) (1) RHS ✓

(iii) (2) 4:9 ✓

32

Prove that : $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$

⇒ LHS side

$$\Rightarrow \frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}}$$

$$= \frac{\frac{\sin \theta}{\cos \theta}}{\frac{\sin \theta - \cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{\frac{\cos \theta - \sin \theta}{\cos \theta}}$$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} \times \frac{\sin \theta}{\sin \theta - \cos \theta} + \frac{\cos \theta}{\sin \theta} \times \frac{\cos \theta}{\sin \theta - \cos \theta}$$

$$= \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{-\sin \theta (\sin \theta - \cos \theta)}$$

$$= \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} - \frac{\cos^2 \theta}{\sin \theta (\sin \theta - \cos \theta)}$$

$$= \frac{\sin^3 \theta - \cos^3 \theta}{\cos \theta \sin \theta (\sin \theta - \cos \theta)}$$

$$= \frac{(\sin^2 \theta + \sin \theta + \cos \theta + \cos^2 \theta)}{\cos \theta \sin \theta}$$

$$= \frac{1 + \sin \theta + \cos \theta}{\cos \theta \sin \theta}$$

$$= \frac{1 + \sec \theta \csc \theta}{\cos \theta \sin \theta} \text{ hence proved.}$$

$$\underline{1353, 861}$$

30. using euclid's algorithm

$$1353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

$$492 = 369 \times 1 + 123$$

$$\underline{369 = 123 \times 3 + 0}$$

$$\text{HCF} = \underline{369}$$

30 ~~tan~~ If $\tan A + \sin A = m$, $\tan A - \sin A = n$,